Reflection

I corrected and improved upon AI in order to make sure the code actually functioned as intended. AI would often generate code that did not work with my current project or created major issues, so I made sure to understand what it was showing and made edits to ensure the methods functioned properly. For instance, when generating the botMove() function, Claude's original attempt simply placed the first valid move rather than prioritizing large pieces and central positioning. Furthermore, the bot would make the same move every game, so I made sure to introduce some randomness to the method and prompted Claude to make more intelligent moves.

Knowing how I would implement methods on my own allowed me to better prompt AI because I was able to describe how I wanted a method to work and why errors were occurring. This gave the AI more detail, ensuring it generated code that worked for my project. Furthermore, understanding the computer science concepts the AI used let me break down what it generated, analyze any potential issues, and simply understand how it worked.

Sprite and Inheritance Structure (Gemini)

I am making the board game Blokus in java. I want you to create a parent Sprite class that the grid and pieces use. Essentially, the Sprite class would have a method to draw a square, and the grid and piece classes would use it. Specifially, the piece objects would refer to any pieces not placed, and the grid object would refer to the board featuring all placed pieces.

Do not use protected

let's look at the piece's x and y position how are they currently calculated and used

ok there should be a set for x and y in the parent class then, and the pieces need to be able to rotate

also add rotate counterclockwise please

let's add a collision detection for a point. check if x, y is in one of the squares and return true or false

the grid's information (the 2d array is supposed to be in another class), can you make it so draw takes in a 2d array rather than using an instance variable

how does this impact piece

ok i am going to keep the board data (2d array) using ints to represent different colors. can you make grid take in a 2d integer array (board) and an int array (piece colors). The int in the 2d array is the index in the 1d array for the color used

i added a boolean instance variable selected to piece with get and set for movement (just letting you know)

piece movement and rotation works now but there is an issue. when pieces are rotated and then moved they kind of teleport. Essentially, selecting the piece works the same, but moving the piece can jump it significantly. Is it possible to add a move function that takes in delta x and delta y and uses it as the amount to move

Right now, pieces can be rotated as if they were layed flat on a table and could only be moved in 2d. This prevents pieces from being flipped into mirrored versions that would occur in the physical game by moving the piece in the third dimension. can you add a way to mirror the pieces horizontally and vertically. Im thinking two seperate methods for horizontal and vertical that take in a boolean flipped. The piece class would have an instance variable flipped and if this.flipped != flipped, this.flipped = flipped and the piece is mirrored

Two questions: why isnt there a flipHorizontal and flipVertical and is it good practice to have your boolean getters be is as opposed to get

Bot Moves (Claude)

This is my java project for the board game blokus. Can you write a botMove() method that makes a valid, legal move for the current player (playerNum)

ok prioritizing big pieces is good, but it should also prioritize central positioning as opposed to the first position that is availibe

i want the piece to move towards the correct position make it so once a piece is selected as the best, its position isnt changed but its targetX and targetY are changed. also do not move to the next player in bot move

actually, the bot move should calculate an X speed and Y speed that will make the piece move 5 pixels in the exact direction of the correct position

what should i change in botMove

what is botTargetRow/Col

how do i make the pieces faster

where should i check if the first player is a bot because right now the first move isnt botted but everyting else is

how do i make the passPlay() repaint so the piece aligns at the end

no at the very end align works perfectly until the final move i just want to call repaint in pass play but that throws an error

ok calling repaint in passPlay did not work can you figure out why the very last play before the endscreen isnt shown autoaligned during the pause

can you add some random ness to bot move because currently, the same exact game happens every time. im thinking you shuffle the pieces in their respective subgroups. that is, the pieces are sorted by size in in each sub group of size, they are shuffled

there is another error with the very fianl piece playerd where draw() attempts to draw it but its been removed by update

no the issue is in the for loop that draws pieces

it is indexout of bounds